

FACULTY OF BASIC MEDICAL SCIENCES

MBBS 214: Human Physiology II - Second Semester Examination 2024

Lecturer: Dr. A. Okafor Time allowed: 2 hours Total marks: 40

Instructions: Answer ALL questions in Sections A and B. Answer ONE question in Section C.

SECTION A: Multiple Choice (10 marks). Each question carries 1 mark.

1. The volume of blood remaining in the ventricle after ejection is the:

- A. End-diastolic volume
- B. End-systolic volume
- C. Stroke volume
- D. Cardiac reserve

2. Which valve closes at the start of ventricular diastole?

- A. Mitral valve
- B. Tricuspid valve
- C. Aortic valve
- D. Both atrioventricular valves

3. Resting cardiac output in a healthy adult is approximately:

- A. 1 L/min
- B. 5 L/min
- C. 15 L/min
- D. 25 L/min

4. Renal plasma flow is best estimated by the clearance of:

- A. Inulin
- B. Creatinine
- C. Para-aminohippuric acid (PAH)
- D. Glucose

5. If the clearance of a substance is greater than GFR, the substance is:

- A. Reabsorbed by the tubules
- B. Secreted by the tubules
- C. Not filtered
- D. Metabolised by the kidney

6. Filtration fraction is the ratio of:

- A. GFR to renal plasma flow
- B. GFR to renal blood flow
- C. Urine flow to GFR
- D. Renal plasma flow to renal blood flow

7. The resting membrane potential of a typical neuron is closest to:

- A. +30 mV
- B. 0 mV
- C. -70 mV

D. -120 mV

8. Myelination increases conduction velocity by:

- A. Increasing axon diameter
- B. Allowing saltatory conduction between nodes of Ranvier
- C. Opening more K⁺ channels
- D. Reducing Na⁺/K⁺ ATPase activity

9. A tidal volume of 500 mL and a respiratory rate of 12 breaths per minute gives a minute ventilation of:

- A. 600 mL/min
- B. 4200 mL/min
- C. 6000 mL/min
- D. 6500 mL/min

10. Anatomical dead space in a healthy adult is approximately:

- A. 50 mL
- B. 150 mL
- C. 500 mL
- D. 1200 mL

SECTION B: Short Answer (20 marks). Each question carries 5 marks.

11. Define ejection fraction and calculate it when end-diastolic volume is 120 mL and end-systolic volume is 50 mL. (5 marks)

12. Explain why glucose appears in the urine in uncontrolled diabetes mellitus, with reference to the renal threshold. (5 marks)

13. Describe how a nerve action potential is propagated along a myelinated axon. (5 marks)

14. Differentiate between obstructive and restrictive lung disease using the FEV₁/FVC ratio and vital capacity. (5 marks)

SECTION C: Theory (10 marks). Answer ONE question.

15. Explain the ionic basis of the nerve action potential, including threshold, the all-or-none principle and the refractory periods. (10 marks)

16. Describe the lung volumes and capacities, and explain how alveolar ventilation differs from minute ventilation. Calculate the alveolar ventilation if tidal volume is 500 mL, dead space is 150 mL and respiratory rate is 12 breaths per minute. (10 marks)